Prioritizing Sustainable Protein Sources with Marine Aquaculture

EDS223 HW3

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Background - DELETE THIS

For this assignment, you are tasked with determining which Exclusive Economic Zones (EEZ) on the West Coast of the US are best suited to developing marine aquaculture for several species of oysters and a species of your choice. Suitable locations will be determined based on range of suitable sea surface temperature (SST) and depth values for the species.

To make your workflow generalizable, you must create a function that has the following characteristics:

```
arguments:
    minimum and maximum sea surface temperature
    minimum and maximum depth
    species name
outputs:
    map of EEZ regions colored by amount of suitable area
    species name should be included in the map's title
```

Tip: Building a generalizable workflow

```
Remember, functions can be hard to debug! So it's highly recommended that you build the worki

Questions to ask yourself: which variables can stay the same for each species and which ones
```

Prepare data

```
# Load packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(here)
```

```
here() starts at /Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/Aquacultu
```

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(tmap)
```

```
Registered S3 method overwritten by 'stars':
  method                   from
  st_interpolate_aw.stars sf
```

```
library(terra)
```

```
terra 1.9.50

Attaching package: 'terra'

The following object is masked from 'package:tidyr':

    extract
```

```
library(knitr)
```

```
Attaching package: 'knitr'

The following object is masked from 'package:terra':

    spin
```

```
#library(ggpubr)

# Set warnings to display as they occur
options(warn = 1)

# Set ggplot theme
set_theme(theme_bw())

# Set here path
here::set_here("/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/Aquacultu
```

```
File .here already exists in /Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds
```

```
# LOAD DATA

# Load west coast EEZ vector file
eez <- sf::st_read(here("data","wc_regions_clean.shp")) #|>
```

```
Reading layer `wc_regions_clean' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/Aquaculture_Potential_
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

```
        # Correct invalid geometries to avoid error in st_join when using CRS 4326
        #st_make_valid()
```

```r
# Load bathymetry raster
bath <- rast(here("data","depth.tif"))

# Load sea surface temp (sst) rasters
sst_08 <- rast(here("data","average_annual_sst_2008.tif"))
sst_09 <- rast(here("data","average_annual_sst_2009.tif"))
sst_10 <- rast(here("data","average_annual_sst_2010.tif"))
sst_11 <- rast(here("data","average_annual_sst_2011.tif"))
sst_12 <- rast(here("data","average_annual_sst_2012.tif"))

# Stack sst rasters
sst_stack <- c(sst_08,sst_09,sst_10,sst_11,sst_12)

# Check resolution and extent of sst_stack
cat("=== Sea Surface Temp Stack (K) ===\n")
```

```
=== Sea Surface Temp Stack (K) ===
```

```r
cat("Resolution:", res(sst_stack), "\n")
```

```
Resolution: 0.04166185 0.04165702
```

```r
cat("Extent:    "); print(ext(sst_stack))
```

```
Extent:
```

```
SpatExtent : -131.98475232999999, -114.98671702755556, 29.993052252666669, 49.98842296400000
```

```r
# Check resolution and extent of bath
cat("=== Bathymetry (m) ===\n")
```

```
=== Bathymetry (m) ===
```

```r
cat("Resolution:", res(bath), "\n")
```

```
Resolution: 0.004166667 0.004166667
```

```r
cat("Extent:    "); print(ext(bath))
```

```
Extent:
```

```
SpatExtent : -132, -114, 28.999999999999996, 50 (xmin, xmax, ymin, ymax)
```

```r
# Warning - CRS - confirm all three crs match, otherwise stop
if (same.crs(eez,bath) & same.crs(sst_stack, eez)){
    # All three CRS match - true
    print("SST Stack, West Coast EEZ, and Bathymetry data CRS all match!")
} else {
    # CRS match - false
    stop("CRS do NOT match!")
}
```

```
[1] "SST Stack, West Coast EEZ, and Bathymetry data CRS all match!"
```

```r
# # Confirm same CRS before proceeding
# cat("Check CRS Match:\n")
# cat("West Coast EEZ vs bathymetry: ", same.crs(eez, bath), "\n")
# cat("SST Stack vs West Coast EEZ: ", same.crs(sst_stack, eez), "\n")

# All CRS match, but EPSG code is missing from sst_stack.
# Assign ESPG:4326 code to sst_stack
crs(sst_stack) <- "EPSG:4326"
```

Process data

```r
# Create raster with 1-layer of mean SST degC from 5-layer raster
sst_mean <- app(sst_stack, fun = "mean", na.rm = TRUE)
names(sst_mean) <- "Mean_SST"

# Check input and output raster layers to confirm
cat("Original SST stacked raster has:",nlyr(sst_stack),"layers","\n")
```

```
Original SST stacked raster has: 5 layers
```

```r
cat("New SST Mean raster has:",nlyr(sst_mean),"layer","\n")
```

```
New SST Mean raster has: 1 layer
```

```r
# Check units of SST_mean raster
minmax(sst_mean)
```

```
    Mean_SST
min  278.130
max  306.045
```

```r
# Convert SST_mean K to C (local operation - each cell)
sst_mean_C <- sst_mean - 273.15

# Confirm units of SST_mean_C raster
minmax(sst_mean_C)
```

```
    Mean_SST
min    4.980
max   32.895
```

```r
# Crop the bath raster to the extent of the sst_mean_C raster
# Note: CRS must match
bath_crop <- crop(bath, sst_mean_C)
```

```r
# Resample bath to match sst_mean_C using bilinear approach
bath_resampled <- resample(bath_crop, sst_mean_C, method = "bilinear")
```

```r
# Warning - confirm bathymetry and SST raster match before proceeding
# Note: compare crs, resolution, and extent
print("Checking that sst_mean_C and bath_resampled rasters match in CRS, resolution, and ext
```

```
[1] "Checking that sst_mean_C and bath_resampled rasters match in CRS, resolution, and exten
```

```r
if (same.crs(bath_resampled,sst_mean_C)){
    # CRS match - true
    print("CRS do match!")
    if (all(res(sst_mean_C) == res(bath_resampled))){
```

```
        # Resolution match - true
        print("Resolutions do match!")
        if (ext(sst_mean_C) == ext(bath_resampled)) {
            # Extent match - true
            print("Extents do match!")
            print("Success - sst_mean_C and bath_resampled match in all required raster chara
        } else {
            # Extent match - false
            stop("Extents do NOT match!")
        }
    } else {
        # Resolution match - false
        stop("Resolutions do NOT match!")
    }
} else {
    # CRS match - false
    stop("CRS do NOT match!")
}
```

```
[1] "CRS do match!"
[1] "Resolutions do match!"
[1] "Extents do match!"
[1] "Success - sst_mean_C and bath_resampled match in all required raster characteristics."
```

```
# Stack sst_mean_C and bath_resampled rasters
ocean_raster <- c(sst_mean_C,bath_resampled)
# Confirm stack
cat("New ocean_raster has",nlyr(ocean_raster),"layers.\n")
```

```
New ocean_raster has 2 layers.
```

```
cat("Layers are:",names(ocean_raster)[nlyr(ocean_raster)-1],",",names(ocean_raster)[nlyr(ocea
```

```
Layers are: Mean_SST , depth
```

Find suitable locations

```r
# Set matrix for classify categories - tempC
rcl_matrix_tempC <- matrix(c(
   11,  30,  1
), ncol = 3, byrow = TRUE)

# Reclassify tempC layer and turn resulting categories into factors
oyster_tempC_zones <- classify(ocean_raster[[1]], rcl = rcl_matrix_tempC, others = 0)

# Define level names
levels(oyster_tempC_zones) <- data.frame(
  value = 0:1,
  label = c("Unsuitable", "Suitable")
)

# Set matrix for classify categories - depthm
rcl_matrix_depthm <- matrix(c(
   -70,  0,  1
), ncol = 3, byrow = TRUE)

# Reclassify depthm layer and turn resulting categories into factors
oyster_depthm_zones <- classify(ocean_raster[[2]], rcl = rcl_matrix_depthm, others = 0)

# Define level names
levels(oyster_depthm_zones) <- data.frame(
  value = 0:1,
  label = c("Unsuitable", "Suitable")
)

# Multiply layers to classify by both tempC and depthm
# Note: Only 1*1 cells will stay 1
habitat_class <- oyster_tempC_zones * oyster_depthm_zones

# Rename label
names(habitat_class) <- "Suitable_Habitat"
```

Determine the most suitable EEZ

```r
# Turn eez vector data into raster
eez_raster <- rasterize(eez, habitat_class, field = "rgn")
```

```
# Confirm rasterization of eez matches habitat_class raster
# Warning - confirm eez_raster and SST raster match before proceeding
# Note: compare crs, resolution, and extent
print("Checking that habitat_class and eez_raster rasters match in CRS, resolution, and exte
```

```
[1] "Checking that habitat_class and eez_raster rasters match in CRS, resolution, and extent
```

```
if (same.crs(habitat_class,eez_raster)){
    # CRS match - true
    print("CRS do match!")
    if (all(res(habitat_class) == res(eez_raster))){
        # Resolution match - true
        print("Resolutions do match!")
        if (ext(habitat_class) == ext(eez_raster)) {
            # Extent match - true
            print("Extents do match!")
            print("Success - habitat_class and eez_raster match in all required raster chara
        } else {
            # Extent match - false
            stop("Extents do NOT match!")
        }
    } else {
        # Resolution match - false
        stop("Resolutions do NOT match!")
    }
} else {
    # CRS match - false
    stop("CRS do NOT match!")
}
```

```
[1] "CRS do match!"
[1] "Resolutions do match!"
[1] "Extents do match!"
[1] "Success - habitat_class and eez_raster match in all required raster characteristics."
```

```
# Grid cell area needs to be determined since crs is EPSG:4326 (units = deg)
# Note: m^2 is default unit
habclass_cellarea <- cellSize(habitat_class, unit = "km")

# Multiply each cell in habitat_class by the cellSize areas
```

```r
# Note: Because unsuitable = 0, unsuitable cells area will = 0
# Note: Only suitable cells (1) will keep the size value (m^2)
habitat_class_area <- habitat_class * habclass_cellarea

# Rename to add units
names(habitat_class_area) <- "Suitable_Habitat_km2"
```

```r
# Calculate sum of suitable area in each EEZ using zonal operation
# habitat_class_area = raster where every suitable pixel represents size in m^2
# eez_raster = raster version of EEZ vector (EEZ regions)
eez_priority <- zonal(habitat_class_area, eez_raster, fun = "sum", na.rm = TRUE)

# Sort eez regions by descending area
eez_priority <- eez_priority |>
    arrange(desc(Suitable_Habitat_km2))
```

```r
# GENERATE FIGURE

# Set color palette
color_palette <- c("#d9d9d9", "#0077b6")

# Add back EEZ polygon geometries for plotting
eez_mapv <- eez |>
    left_join(eez_priority, by = "rgn") |>
    # Sort by desc. suitable area
    arrange(desc(Suitable_Habitat_km2))


# Create EEZ region map colored by amount of suitable area
# Add basemap
tm_basemap(server = "Esri.WorldTopoMap") +
# Add EEZ polygons
tm_shape(eez_mapv) +
  tm_polygons(fill = "Suitable_Habitat_km2",
              fill_alpha = 0.6, # Transparency
              col = "#36454F", # Border color
              # Define color palette for 'aquaculture suitability' fill
              fill.scale = tm_scale_continuous(values = color_palette),
              # Re-label fill legend and adjust position
              fill.legend = tm_legend(title = "Suitable Area (km^2)",
                                      position = tm_pos_out("right"))) +
  # Add region key as text inside each EEZ polygon
```

```
  tm_text(text = "rgn_key",
          size = 0.8,
          col = "#36454F") +
# Add compass and scalebar
tm_compass(position = c("left","bottom")) +
tm_scalebar(position = c("left","bottom")) +
# Adjust pdf figure sizing - turn off autoscale
tm_options(component.autoscale = FALSE) +
# Add figure title
tm_title("EEZ Regions with the Most Suitable Area for Oyster Aquaculture",
            size = 1.5,
            frame = TRUE) +
# Adjust legend text, fonts, and margins
tm_layout(# Specify legend title size
          legend.title.size = 1.0,
          # Specify legend text size
          legend.text.size = 0.8,
          # Set font family
          text.fontfamily = "sans",
          # Set inner margins to give scalebar room
          inner.margins = c(0.1,0.1,0.05,0.05)
          )
```

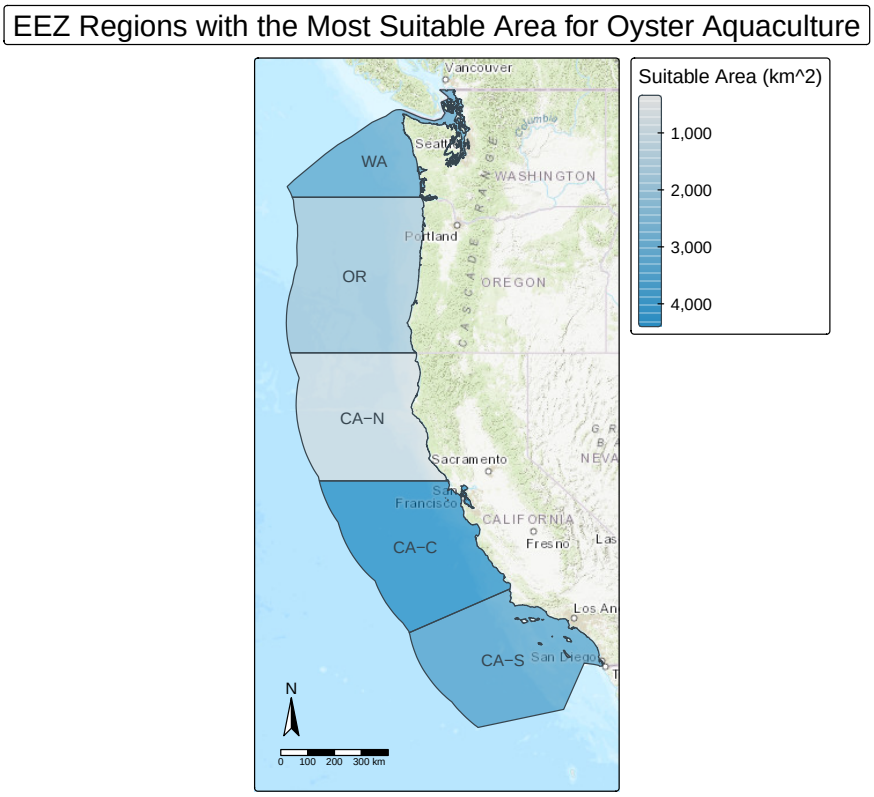


Figure 1: Map of the US West Coast Exclusive Economic Zones (EEZ), colored by amount of suitable area (km^2) for oyster marine aquaculture. Suitable area for oyster aquaculture is defined by areas meeting both the temperature requirement of 11-30 degree C, and the depth requirement of 0 to 70 m.

```
# Make table using kable
eez_mapv |>
    select(rgn, rgn_key, Suitable_Habitat_km2) |>
    st_drop_geometry() |>
    knitr::kable(col.names = c("EEZ Region", "Region Key", "Total Suitable Habitat (km^2)"))
```

Table 1: Total suitable area (km^2) for oyster marine aquaculture in each US West Coast Exclusive Economic Zone (EEZ). Suitable area for oyster aquaculture is defined by areas meeting both the temperature requirement of 11-30 degree C, and the depth requirement of 0 to 70 m.

EEZ Region	Region Key	Total Suitable Habitat (km^2)
Central California	CA-C	4373.5565

EEZ Region	Region Key	Total Suitable Habitat (km^2)
Southern California	CA-S	3329.9002
Washington	WA	3091.5158
Oregon	OR	1395.7328
Northern California	CA-N	323.9124

Data

Flanders Marine Institute (2026): MarineRegions.org. https://www.marineregions.org/eez.php

GEBCO Compilation Group. (2022). GEBCO_2022 Grid [Data set]. British Oceanographic Data Centre, National Oceanography Centre, NERC, UK. https://doi.org/10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c

NOAA Coral Reef Watch. (2018). Daily global 5km satellite sea surface temperature anomaly (Version 3.1). U.S. Department of Commerce, National Oceanic and Atmospheric Administration. https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php

References

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